

Multiple systems

Probabilistic states

correlation: it is a lack of independence.

Tensor Product

$$\text{Given: } |\phi\rangle = \sum_{a \in \Sigma} \alpha_a |a\rangle \quad \& \quad |\psi\rangle = \sum_{b \in \Gamma} \beta_b |b\rangle$$

$$\text{Tensor Product } |\phi\rangle \otimes |\psi\rangle = \sum_{(a,b) \in \Sigma \times \Gamma} \alpha_a \beta_b |ab\rangle$$

$$|\pi\rangle = |\phi\rangle \otimes |\psi\rangle$$

$$\langle ab | \pi \rangle = \langle a | \phi \rangle \langle b | \psi \rangle$$

For a joint system (X, Y) in a probabilistic state represented by a probability vector $|\pi\rangle$, the systems X & Y are independent if $|\pi\rangle$ is obtained by taking a tensor product

$$|\pi\rangle = |\phi\rangle \otimes |\psi\rangle$$

$|\pi\rangle$ is then a product state/vector

notation:

$|\phi\rangle \otimes |\psi\rangle$ is same as writing $|\phi\rangle |\psi\rangle$

$|\phi \otimes \psi\rangle$ also sometimes used.

$$\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix} \otimes \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} = \begin{pmatrix} \alpha_1 \beta_1 \\ \vdots \\ \alpha_1 \beta_k \\ \alpha_2 \beta_1 \\ \vdots \\ \alpha_m \beta_1 \\ \vdots \\ \alpha_m \beta_k \end{pmatrix}$$

Tensor product of two vectors is bilinear.

Bilinear: linear in each of the two arguments separately, assuming that the other argument is fixed.

1. Linearity in first argument:

$$(|\phi_1\rangle + |\phi_2\rangle) \otimes |\psi\rangle = |\phi_1\rangle \otimes |\psi\rangle + |\phi_2\rangle \otimes |\psi\rangle$$

$$(\alpha|\phi\rangle) \otimes |\psi\rangle = \alpha(|\phi\rangle \otimes |\psi\rangle)$$

2. Linearity in second argument:

$$|\phi\rangle \otimes (|\psi_1\rangle + |\psi_2\rangle) = |\phi\rangle \otimes |\psi_1\rangle + |\phi\rangle \otimes |\psi_2\rangle$$

$$|\phi\rangle \otimes (\alpha|\psi\rangle) = \alpha(|\phi\rangle \otimes |\psi\rangle)$$

$$(\alpha|\phi\rangle) \otimes |\psi\rangle = |\phi\rangle \otimes (\alpha|\psi\rangle) = \alpha(|\phi\rangle \otimes |\psi\rangle)$$

Scalars "float freely"

$$\alpha|\phi\rangle \otimes |\psi\rangle \text{ same as } \alpha|\phi\rangle|\psi\rangle \text{ same as } \alpha|\phi \otimes \psi\rangle$$

Tensor product of 3 or more vectors is multilinear

Measurements of probabilistic state

When just X is measured

$$\Pr(X=a) = \sum_{b \in \Gamma} \Pr((X, Y) = (a, b))$$

Conditional probability formula reflects uncertainty remaining on Y

$$\Pr(Y=b|X=a) = \frac{\Pr((X,Y)=(a,b))}{\Pr(X=a)}$$

$$|\pi\rangle = \sum_{(a,b) \in \Sigma \times \Gamma} P_{ab} |ab\rangle$$

measuring X alone: $\Pr(X=a) = \sum_{c \in \Gamma} P_{ac}$

Vector representing probabilistic state of X alone:

$$\sum_{a \in \Sigma} \left(\sum_{c \in \Gamma} P_{ac} \right) |a\rangle$$

probabilistic state of Y :

$$|\nu_a\rangle = \frac{\sum_{b \in \Gamma} P_{ab} |b\rangle}{\sum_{c \in \Gamma} P_{ac}}$$

If X results in state a , probabilistic state of (X,Y) updates to $|a\rangle \otimes |\nu_a\rangle$

Example: X is $\Sigma = \{0,1\}$, Y is $\Gamma = \{1,2,3\}$

Probabilistic state of (X,Y) is

$$|\pi\rangle = \frac{1}{2} |0,1\rangle + \frac{1}{12} |0,3\rangle + \frac{1}{12} |1,1\rangle + \frac{1}{6} |1,2\rangle + \frac{1}{6} |1,3\rangle$$

$$|\pi\rangle = |0\rangle \otimes \left(\frac{1}{2}|1\rangle + \frac{1}{2}|3\rangle \right) + |1\rangle \otimes \left(\frac{1}{2}|1\rangle + \frac{1}{6}|2\rangle + \frac{1}{6}|3\rangle \right)$$

$$\Pr(X=0) = \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{7}{12} \quad \Pr(X=1) = \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{6} \right) = \frac{5}{12}$$

$$\frac{7}{12} + \frac{5}{12} = \frac{12}{12} = 1 \checkmark$$

For probabilistic state of Y conditioned on each possible outcome, normalize vectors calculated.

Normalize: divide calculated vectors by associated probabilities calculated.

For $x=0$, Y is

$$\frac{\frac{1}{2}|1\rangle + \frac{1}{2}|3\rangle}{\frac{7}{12}} = \frac{6}{7}|1\rangle + \frac{6}{7}|3\rangle$$

For $x=1$, Y is

$$\frac{\frac{1}{2}|1\rangle + \frac{1}{6}|2\rangle + \frac{1}{6}|3\rangle}{\frac{5}{12}} = \frac{2}{5}|1\rangle + \frac{2}{5}|2\rangle + \frac{2}{5}|3\rangle$$

Operations on Probabilistic States

Controlled-NOT operation: if $x=1$, perform NOT on Y

Matrix for controlled NOT:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Its action on standard basis states:

$$|00\rangle \mapsto |00\rangle$$

$$|01\rangle \mapsto |01\rangle$$

$$|10\rangle \mapsto |11\rangle$$

$$|11\rangle \mapsto |10\rangle$$

If Y was control & X was target bit:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$|00\rangle \mapsto |00\rangle$$

$$|01\rangle \mapsto |11\rangle$$

$$|10\rangle \mapsto |10\rangle$$

$$|11\rangle \mapsto |01\rangle$$

Tensor Product of matrices

Given matrices M & N

$$M = \sum_{a,b \in \Sigma} \alpha_{ab} |a\rangle \langle b|$$

$$N = \sum_{c,d \in \Gamma} \beta_{cd} |c\rangle \langle d|$$

tensor product $M \otimes N$ is the matrix

$$M \otimes N = \sum_{a,b \in \Sigma} \sum_{c,d \in \Gamma} \alpha_{ab} \beta_{cd} |ac\rangle \langle bd|$$

$$\langle ac | M \otimes N | bd \rangle = \langle a | M | b \rangle \langle c | N | d \rangle$$

$$(M \otimes N)(| \psi \rangle \otimes | \chi \rangle) = (M | \psi \rangle) \otimes (N | \chi \rangle)$$

$$\begin{pmatrix} \alpha_{11} & \dots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \dots & \alpha_{mm} \end{pmatrix} \otimes \begin{pmatrix} \beta_{11} & \dots & \beta_{1k} \\ \vdots & \ddots & \vdots \\ \beta_{k1} & \dots & \beta_{kk} \end{pmatrix}$$

$$= \begin{pmatrix} \alpha_{11}\beta_{11} & \dots & \alpha_{11}\beta_{1k} & \dots & \alpha_{1m}\beta_{11} & \dots & \alpha_{1m}\beta_{1k} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \alpha_{11}\beta_{k1} & \dots & \alpha_{11}\beta_{kk} & \dots & \alpha_{1m}\beta_{k1} & \dots & \alpha_{1m}\beta_{kk} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \alpha_{m1}\beta_{11} & \dots & \alpha_{m1}\beta_{1k} & \dots & \alpha_{mm}\beta_{11} & \dots & \alpha_{mm}\beta_{1k} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \alpha_{m1}\beta_{k1} & \dots & \alpha_{m1}\beta_{kk} & \dots & \alpha_{mm}\beta_{k1} & \dots & \alpha_{mm}\beta_{kk} \end{pmatrix}$$

Independent operations

Example: 2 indep. syst.s X & Y, bits

X: oper. 1: If state of bit is 0, do nothing. If 1, flip to 0
w/ prob. $\frac{1}{2}$.
 $\Rightarrow \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix}$

Y: Oper. 2: NOT

If both independently performed on X & Y, Compound syst. (X,Y) matrix is:

$$\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}$$

Doing nothing to one syst. is represented by the identity matrix.